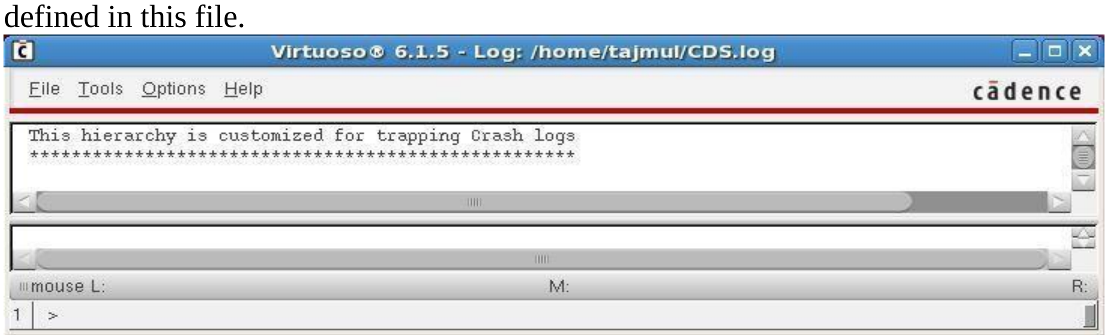
VLSI Design Technology

General Notes:
Fist to perform all the labs we need to install below tools properly:

I. IC615
II. Assura
III. MMSIM
IV. Incisive Enterprise Simulator

Now

1) To start, Log in to your workstation using the username and password.
2) Open terminal and go to the directory where **cshrc** file is located.
   e.g. ***cd /home/license/***
3) In the terminal window, type **csh** at the command prompt to invoke the C shell.
   # csh
4) Source the **cshrc** file.
   # source cshrc
5) Go to the working directory.
   e.g. ***# cd /home/taj/work***
6) Type **virtuoso &** command in terminal.
   # virtuoso &
   It will start the Cadence Design Framework II environment from this directory because it contains cds.lib file, which is the local initialization file. The library search paths are defined in this file.



   This window called CIW (Command Interpreter Window). Keep opened CIW window for the labs.
7) If the ―What's New ...‖ window appears, close it with the **File → Close** command.

Department of Electrical & Electronic Engineering
University of Asia Pacific

EEE424 VLSI I Laboratory
Lab 1: Design a CMOS Inverter.

Objective:
1) Library creation.
2) Cell View creation.
3) Schematic Design.
4) Symbol creation.

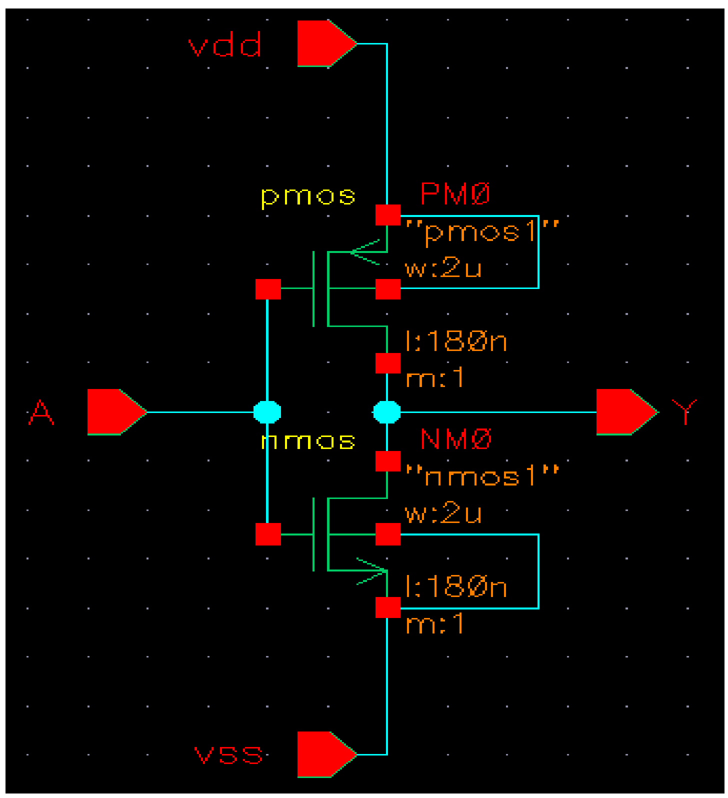


Fig: Schematic view of an Inverter

1) Library creation.

Below steps explain the creation of a new library **―myDesignLib‖** and we will use the same throughout this course for building various cells that we going to create in the next labs. Execute **Tools – Library Manager** in the **CIW (Command Interpreter Window)** or Virtuoso window to open Library Manager.

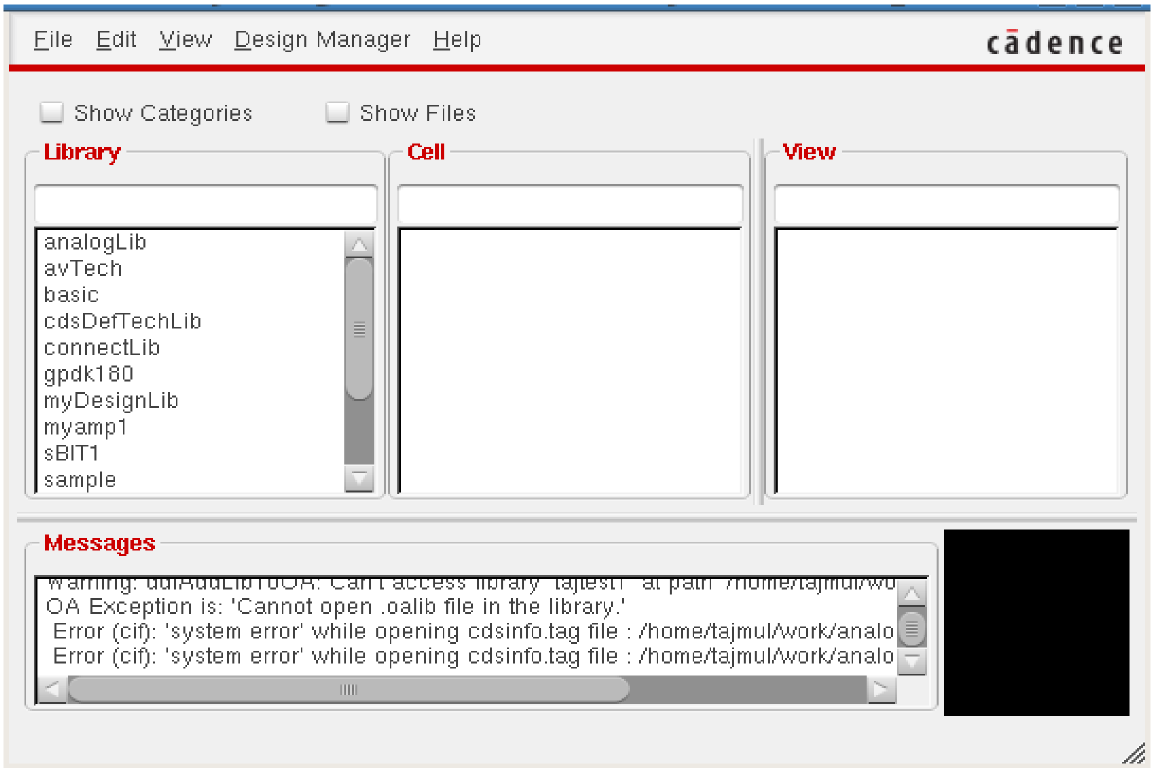
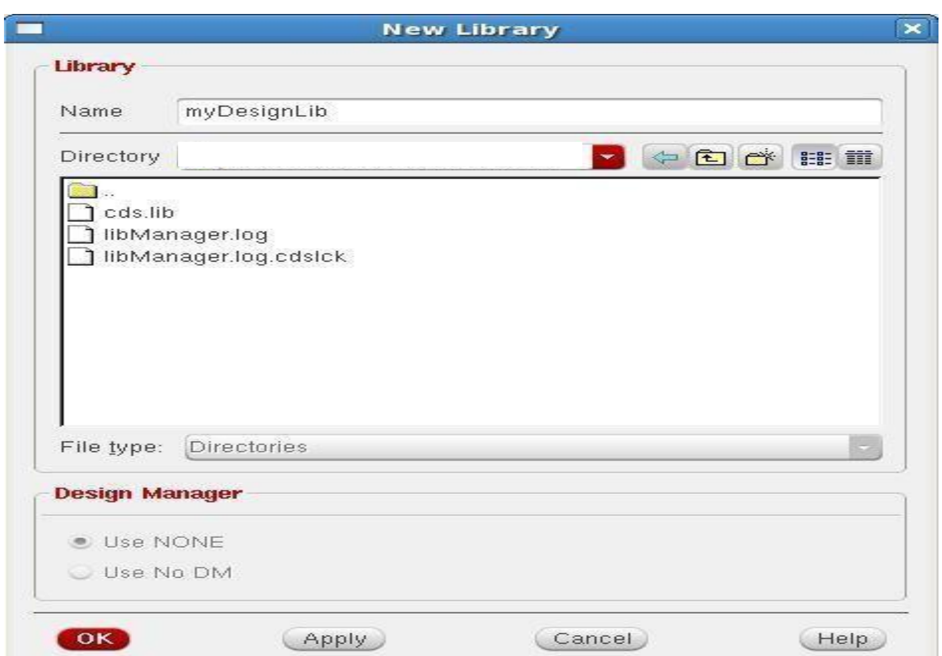


Fig: Library Manager

Creating a New library:

I. In the Library Manager, execute **File - New – Library**. The new library form appears.

II. In the —New Library‖ form, type **—myDesignLib**‖ in the Name section.



III. Fig: New Library windowIn the field of Directory section, verify that the path to the library is set to the current working directory

IV. In the next **—Technology File for New library‖** form pops up, select option **Attach to an existing techfile** and click **OK**.

Fig: Technology File for New library

V. Then **—Attach Design Library to Technology File‖** form pops up, select **gpdk180** from the cyclic field and click **OK**.

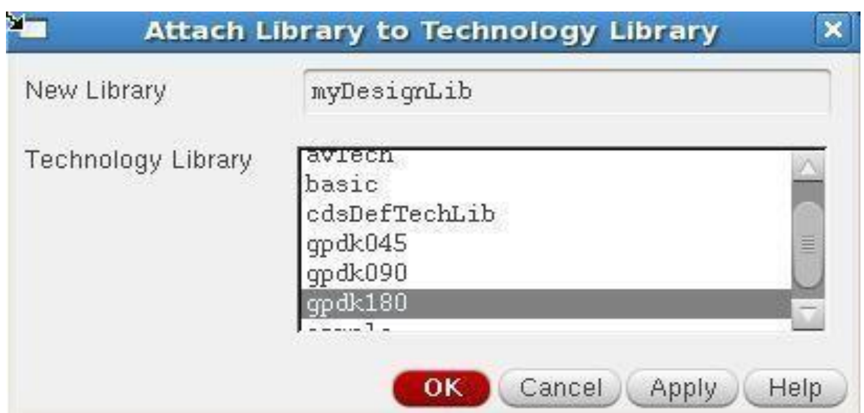
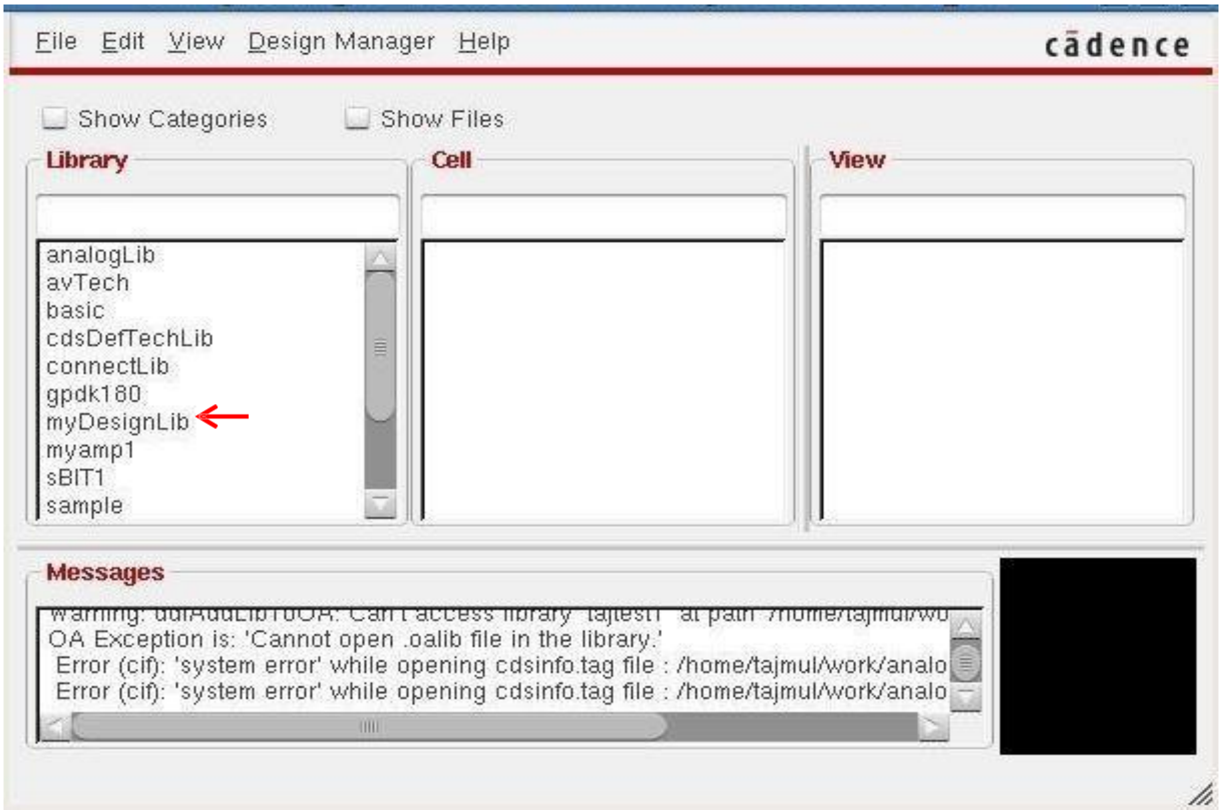


Fig: Attach Design Library to Technology File

VI. After creating a new library we can verify it from the library manager.

VII. If you right click on the **—myDesignLib‖** and select properties, you will find that **gpdk180** library is attached as techlib to **—myDesignLib‖**.



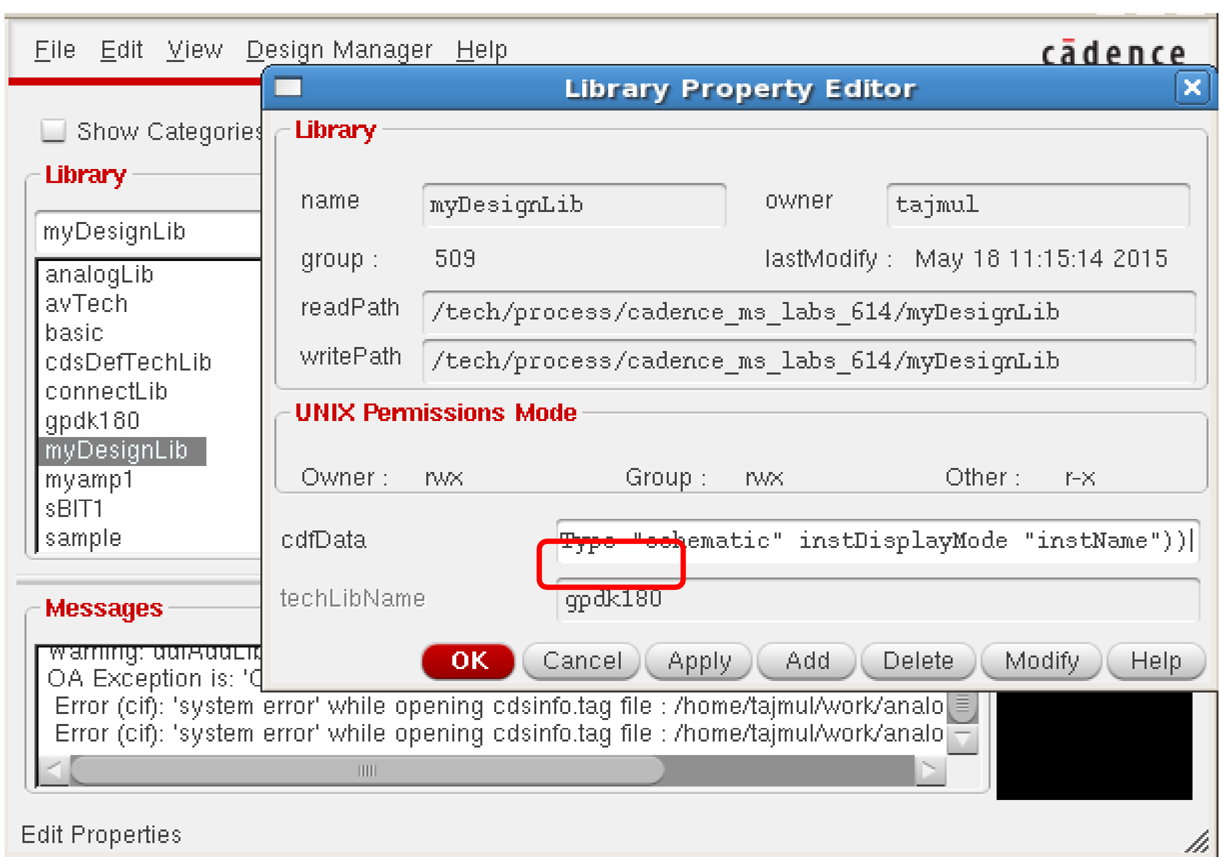
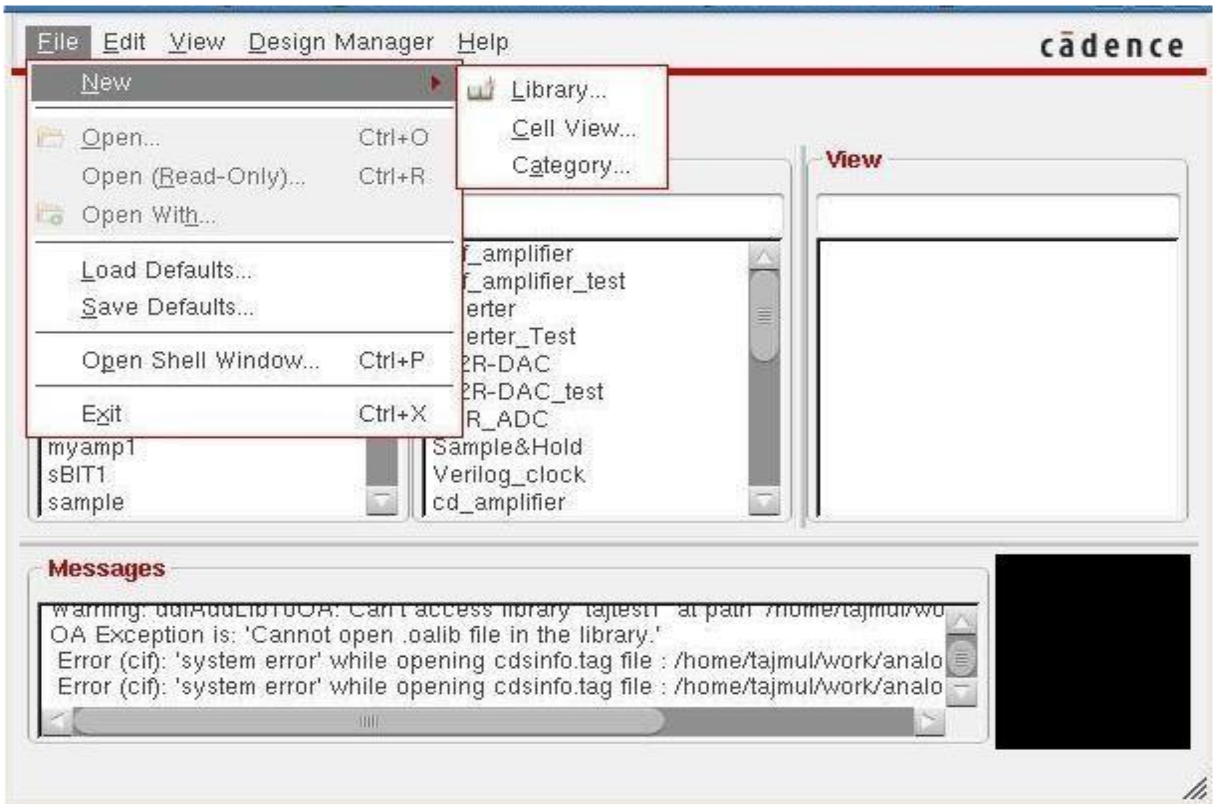


Fig: Library Property

2) Cell View creation:

In this section we will learn how to create a new cell view in the new —**myDesignLib**‖ library.

I. In the CIW or Library manager, execute **File** –> **New** –> **Cellview.**



II. A **New file** form pops up. Set up the **New file** form as follows:

III. In **Open with** option under the **Application** select it as **Schematic XL**.
Do not edit the **Library path file** and the one above might be different from the path shown in your form.

IV. Click **OK** when done the above settings. A blank schematic window for the Inverter design appears.

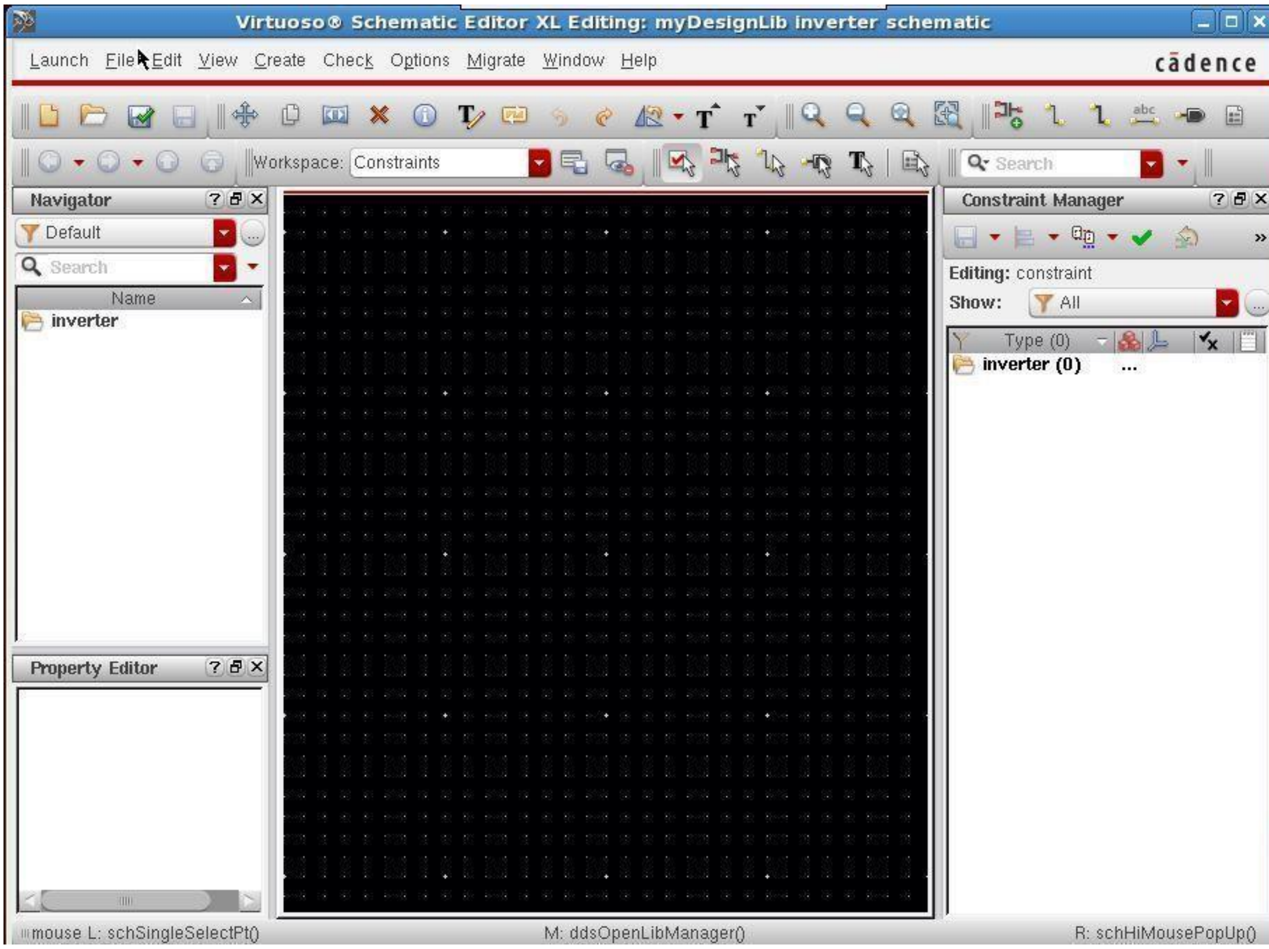
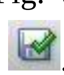


Fig: Virtuoso Schematic Editor

V. Click on **Check and Save** Icon . It will save our cell.

3) Schematic Design.

I. In the Inverter schematic black window click the **Instance** fixed menu Icon  to display the Add Instance form.

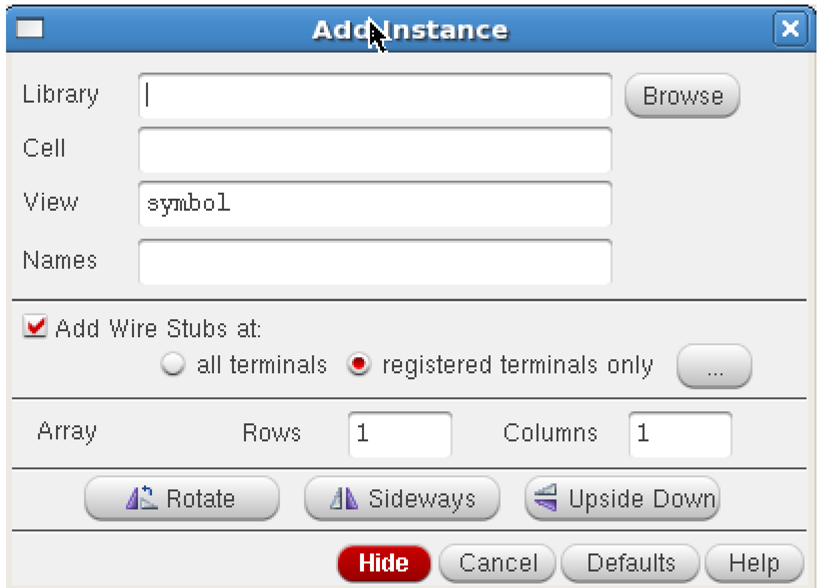


Fig: Add Instance form

Tip: We can also execute Create — Instance or press i.

II. Click on the **Browse** button. This opens up a **Library browser** from which we can select components and the **symbol** view .

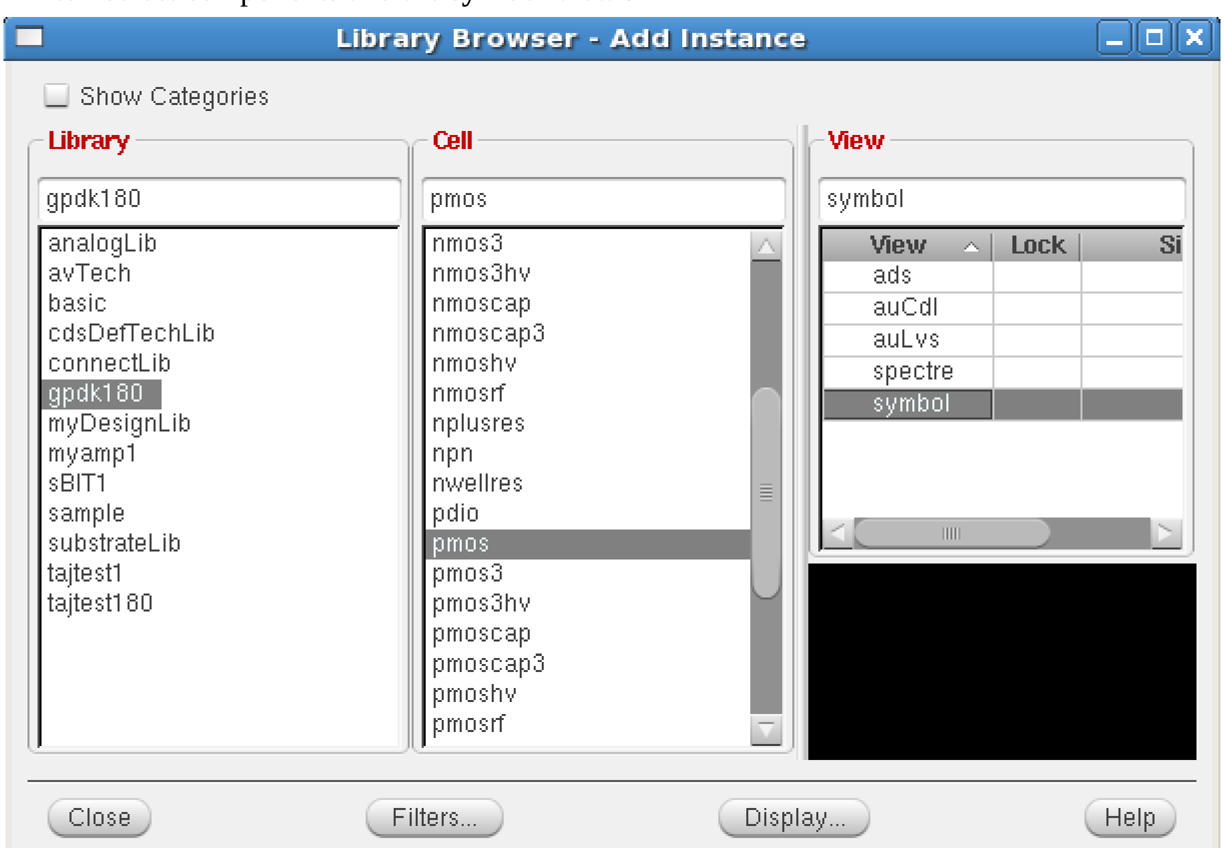
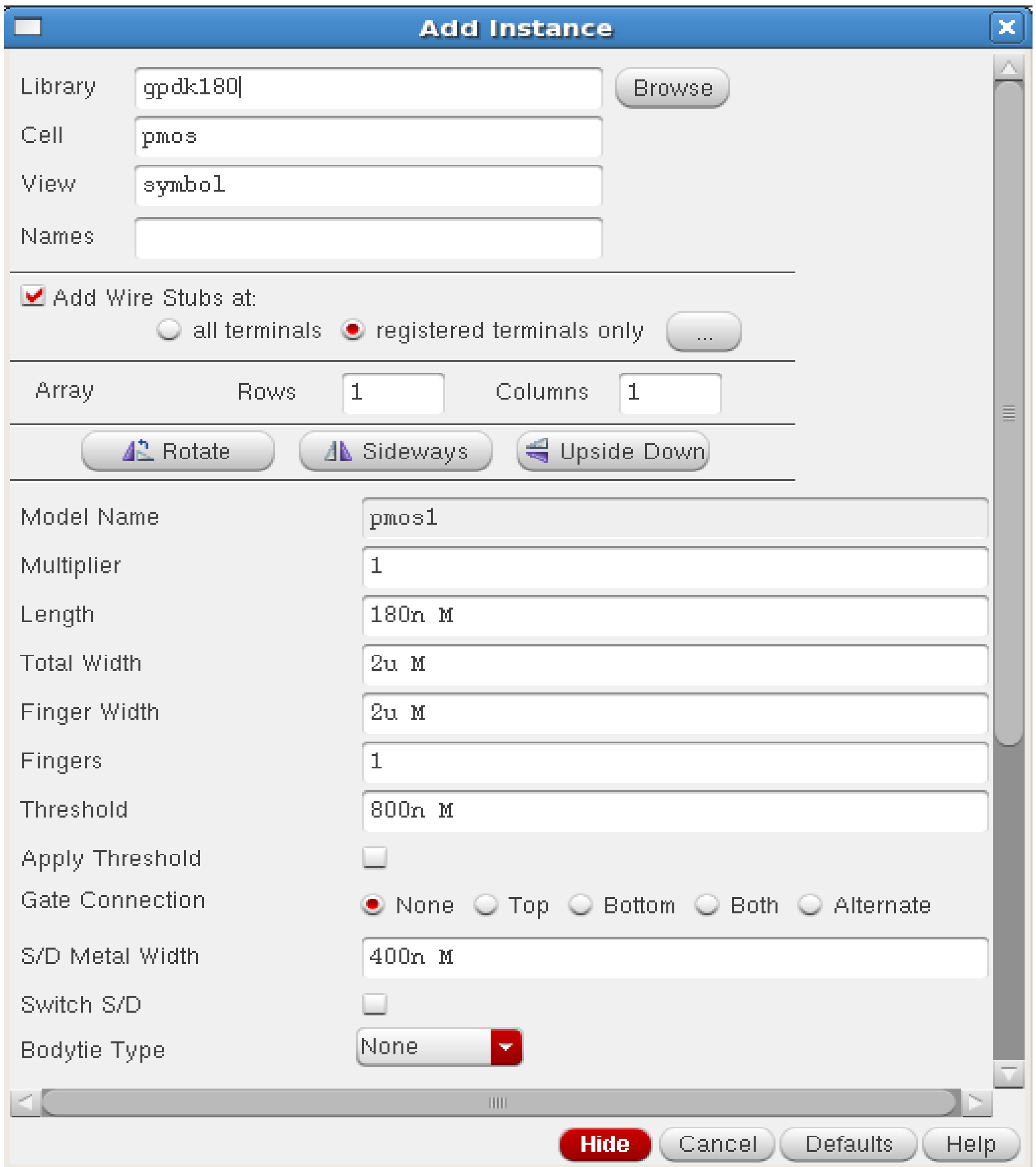


Fig: Library Browser form

III. We will update the Library Name, Cell Name, and the property values given in the table on the next page as you place each component.



IV. After we complete the Add Instance form, move our cursor to the schematic window and click **left** of mouse to place a component.

V. This is a table of components for building the Inverter schematic.

Library name	Cell Name	Properties/Comments
gpdk180	pmos	For M0: Model name = pmos1, W= 4u, L=180n
gpdk180	nmos	For M1: Model name = nmos1, W= 2u, L=180n

If we place a component with the wrong parameter values, use the **Edit— Properties— Objects** command to change the parameters. Use the **Edit— Move** command if you place components in the wrong location.

We can rotate components at the time we place, or use the **Edit— Rotate** command after they are placed.

VI. After entering components, click **Cancel** in the Add Instance form or press **Esc** with your cursor in the schematic window.

VII. Adding pins to Schematic:

1. Click the Pin fixed menu icon  in the schematic window. The Add pin form appears.
   Tip: We can also execute Create — Pin or press p.

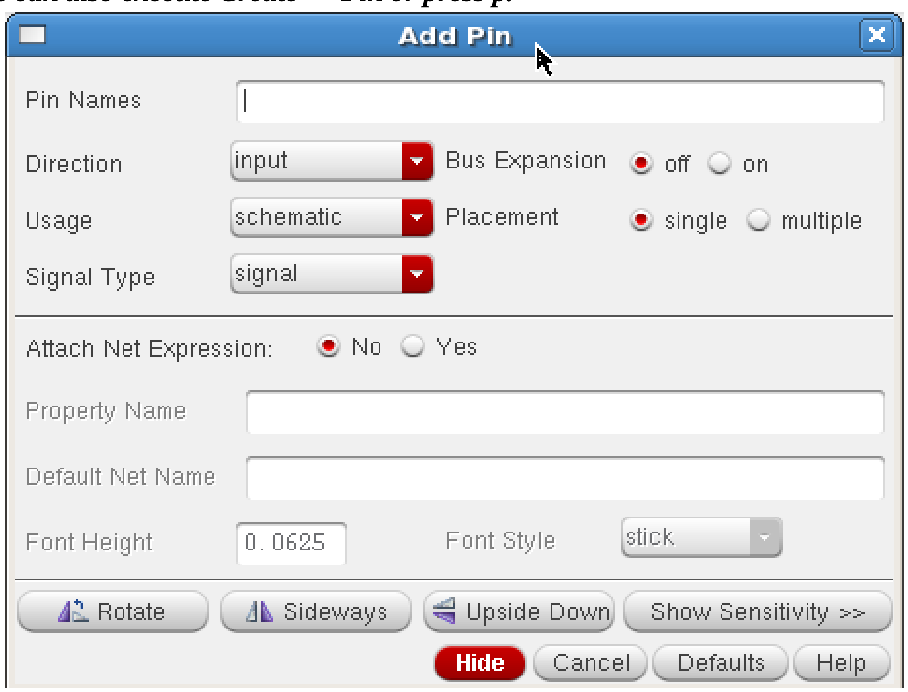


Fig: Add pin form

2. Type the following in the Add pin form in the exact order leaving space between the pin names.

Pin Name	Direction
A	Input
Y	Output
Vdd	Input
Vss	Input

Make sure that the direction field is set to **input/output/inputOutput** when placing the **input/output/inout** pins respectively and the Usage field is set to schematic.

3. Select **Cancel** from the Add – pin form after placing the pins.
   In the schematic window, execute Window— Fit or press the f bindkey.

VIII. Adding Wires to a Schematic:

Add wires to connect components and pins in the design.

1. Click the Wire (narrow) icon  in the schematic window.

We can also press the ***w*** key, or execute **Create — Wire (narrow).**

2. In the schematic window, click on a pin of one of our components as the first point for our wiring. A diamond shape appears over the starting point of this wire.
3. Follow the prompts at the bottom of the design window and click left on the destination point for our wire. A wire is routed between the source and destination points.
4. Complete the wiring as shown in figure and when done wiring press ESC key in the schematic window to cancel wiring.

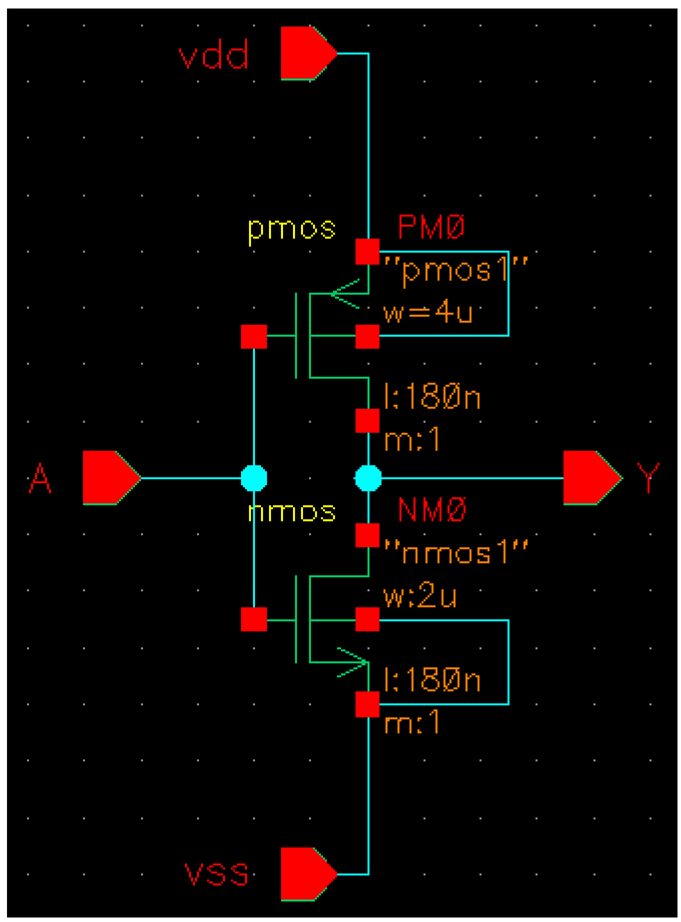


Fig: Inverter schematic view

5. Click the Check and Save icon  in the schematic editor window.
6. Observe the CIW output area for any errors.
7. Now if you check our Library **—myDesignLib‖** in **Library Manager**, we will see the inverter is saved as Schematic view.

4) Symbol creation:

In this section, we will create a symbol for your inverter design so we can place it in a test circuit for simulation. A symbol view is extremely important step in the design process. The symbol view must exist for the schematic to be used in a hierarchy. In addition, the symbol has attached properties that facilitate the simulation and the design of the circuit.

I. In the Inverter schematic window, execute **Create — Cellview— From Cellview**.
The **Cellview From Cellview** form appears. With the Edit Options function active, you can control the appearance of the symbol to generate.

Fig: Cellview From Cellview form

II. Verify that the **From View Name** field is set to **schematic**, and the **To View Name** field is set to **symbol**, with the **Tool/Data Type** set as **SchematicSymbol**.

III. Click **OK** in the **Cellview From Cellview** form. The Symbol Generation Form appears.

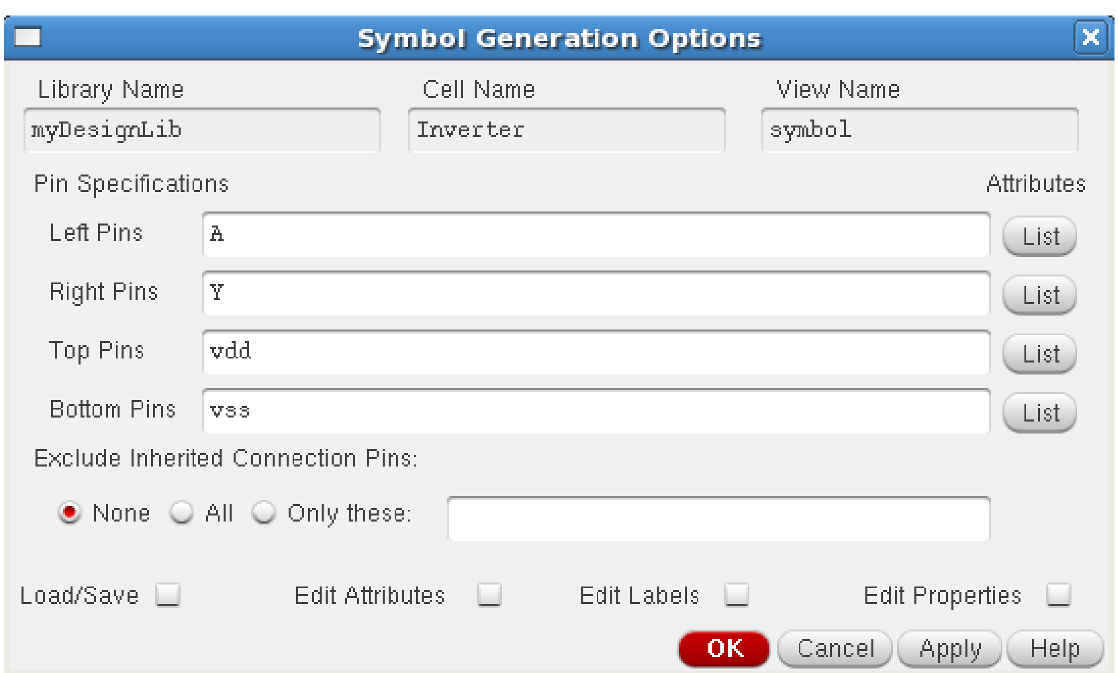
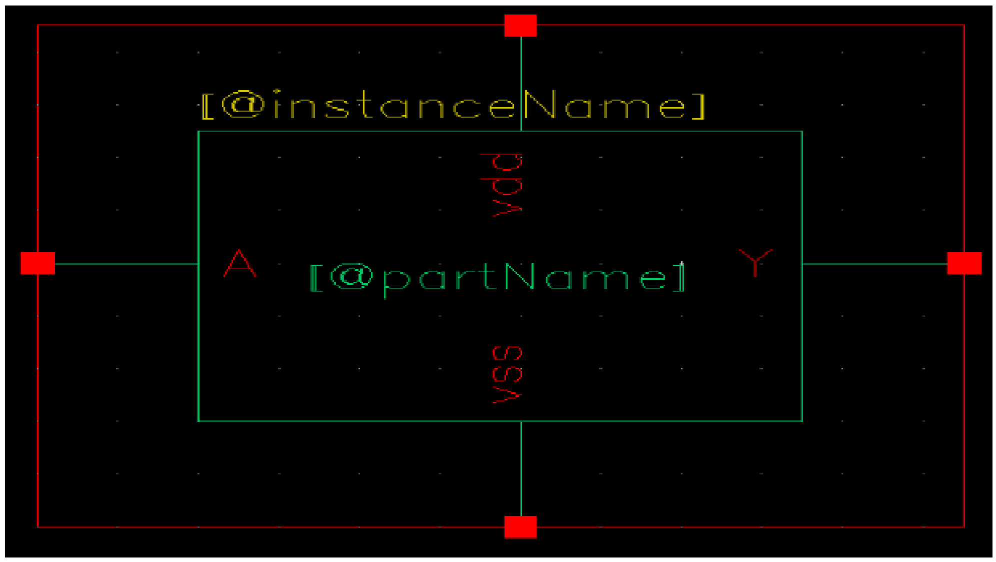


Fig: Symbol Generation Options

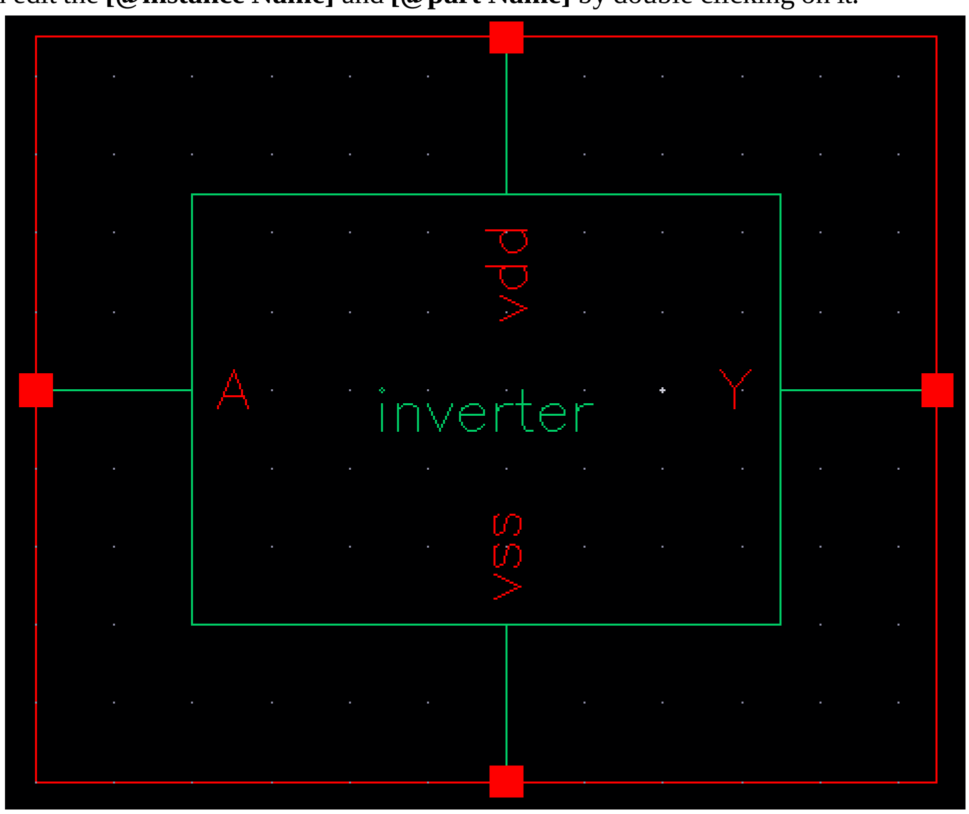
IV. Modify the Pin Specifications as above:

V. Click **OK** in the Symbol Generation Options form.

VI. A new window displays an automatically created Inverter symbol as shown here



VII. We can edit the **[@instance Name]** and **[@part Name]** by double clicking on it.



VIII. Editing the Symbol:

In this section we will modify the inverter symbol to look like a Inverter gate symbol.

1. Move the cursor over the automatically generated symbol, until the green rectangle is highlighted, click **left** to select it.
2. Click **Delete** icon in the symbol window, similarly select the red rectangle and delete that.
3. Execute **Create – Shape – polygon**, and draw a shape similar to triangle.
4. After creating the triangle press **ESC** key.
5. Execute **Create – Shape – Circle** to make a circle at the end of triangle.
6. We can move the pin names according to the location by using M key.
7. Execute **Create — Selection** Box. In the Add Selection Box form, click **Automatic**. A new red selection box is automatically added.
8. After creating symbol, click on the ***save*** icon  in the symbol editor window to save the symbol. In the symbol editor, execute **File — Close** to close the symbol view window.

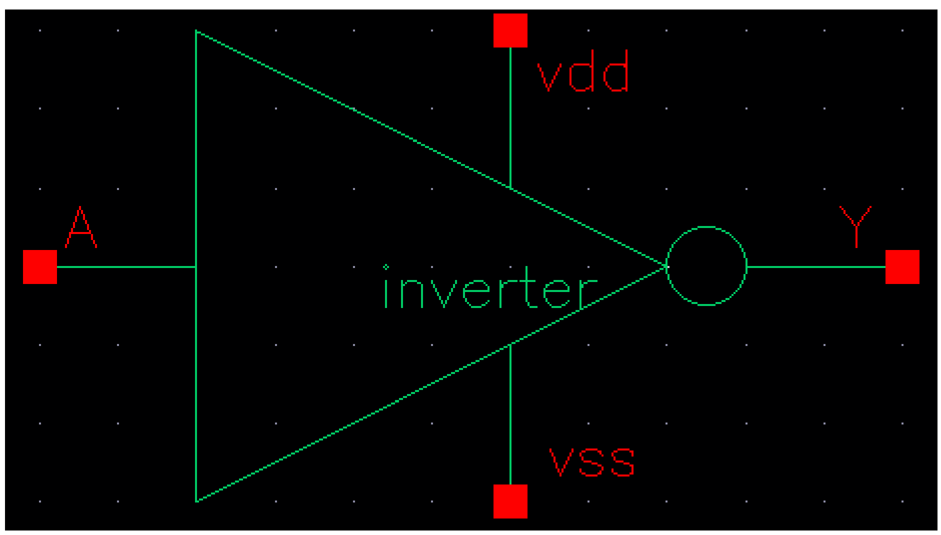


Fig: Inverter Symbol

9. Now if we check our Library **—myDesignLib‖** in **Library Manager**, we will see the inverter is saved as Symbol view.

Report :

- Provide all prelab works.
- Provide all circuit schematics and simulation results.
- Symbol Creation of a two input AND Gate, two input OR Gate and a NOT Gate. Provide Screenshot of Circuit Diagrams and the picture of symbol created.
- Include necessary mathematical expressions and analytical descriptions as required.